

HARSHITHA B

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SUMMARY

M.Tech CSE graduate specializing in **Artificial Intelligence and Machine Learning**, with research experience in **explainable AI for echocardiography**, medical image segmentation, cardiac function analysis, and clinical report generation. Developed an end-to-end **U-Net** pipeline for LV segmentation and LVEF estimation, integrated with **Mistral-7B** for structured clinical report generation. Experienced in Python, PyTorch, TensorFlow, computer vision, and generative AI.

EDUCATION

Vellore Institute of Technology, Vellore, Tamil Nadu

Jul 2024 – Jul 2026

M.Tech – Computer Science & Engineering, AI & ML

CGPA: 8.92/10

TECHNICAL SKILLS

Programming: Python, SQL

Machine Learning: Supervised Learning, Unsupervised Learning, Feature Engineering, Model Evaluation, Random Forest, XGBoost

Deep Learning: PyTorch, TensorFlow, Keras, U-Net, VGG16, ResNeXt, CNN, RNN, DNN

Computer Vision: OpenCV, MTCNN, Haar Cascade, FFmpeg

NLP & Generative AI: Hugging Face Transformers, T5-small, Mistral-7B, Prompt Engineering, Structured LLM Generation

Data & Databases: NumPy, MySQL, SQLite

Cloud & Tools: AWS, GitHub, VS Code, Jupyter Notebook, Kaggle, Flask

EXPERIENCE

Elevate Labs

Aug 2025 – Sept 2025

AIML Intern – Remote

- Completed hands-on AI/ML tasks covering exploratory data analysis, regression, classification, ensemble learning, KNN, SVM, and K-Means clustering using **Python** and **Scikit-learn**.
- Developed an **AI text summarization web application** using **Hugging Face Transformers**, **T5-small**, and **Flask**, generating concise summaries from user-provided text with summary-length statistics.
- Developed a **Flask + SQLite CRUD web application** with HTML/Jinja templates, implementing user creation, retrieval, updating, and deletion; documented work on GitHub and recognized as **Best Performer**.

PROJECTS

Explainable AI for Echocardiography Segmentation & Clinical Report Generation

Python, PyTorch, U-Net, Mistral-7B

- Developed an end-to-end deep learning pipeline for **left-ventricular segmentation and LVEF estimation** from apical four-chamber echocardiography videos.
- Derived end-diastolic/end-systolic phases and LVEF from segmentation-driven LV cavity dynamics, achieving **0.94 Dice**, **0.86 IoU**, **6.16 percentage-point EF MAE**, and $R^2 = 0.613$, with external validation on CAMUS.
- Integrated **Mistral-7B** with controlled prompting to generate structured echocardiographic summaries from verified quantitative measurements.

Deep Learning for Audio-Visual Deepfake Detection

Python, VGG16, Random Forest, FFmpeg, DFDC

- Developed a **multimodal deepfake detection pipeline** combining visual and audio analysis to distinguish authentic and AI-generated media.
- Applied FFmpeg-based audio processing and trained a **Random Forest classifier with 10-fold cross-validation** for synthetic voice detection.
- Fine-tuned a **VGG16-based model** for visual deepfake detection and evaluated performance using accuracy, precision, recall, F1-score, MCC, and ROC-AUC.

Authentic vs AI-Generated Video Detection

Python, ResNeXt, CNN, RNN/DNN, OpenCV

- Developed a deep learning pipeline for authentic vs AI-generated video classification using frame-level visual features and temporal sequence analysis.
- Standardized face-cropped video sequences to **150 frames at 112×112 resolution** and used pretrained **ResNeXt** for frame-level feature extraction followed by a DNN/RNN-based classifier.
- Evaluated diverse datasets including **FaceForensics++**, **DFDC**, **Celeb-DF**, and **YouTube** using classification metrics and confusion-matrix analysis.

PUBLICATIONS

A Robust Deep Learning Approach for Discriminating Between Authentic and AI-Generated Videos – ICETM 2025, Apr. 2025.

Deep Learning for Audio-Visual Deepfake Detection: A Study on Face and Voice Synthesis – ICECA 2025, Nov. 2025.